

SATOSHI KOISO

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OBJECTIVE

To advance evidence-based policymaking using decision science and health economic analyses, improving health outcomes for economically and politically disadvantaged people globally.

Research and Teaching Interests:

Global Health; Health Decision Science; Economic Evaluations; Infectious Diseases; Nutrition; Evidence-based Policymaking; Research Transparency and Reproducibility

PROFESSIONAL EXPERIENCE

Medical Practice Evaluation Center, Massachusetts General Hospital

Boston, MA

Research Specialist (supervised by Dr. Emily Hyle, Dr. Kenneth Freedberg, and Dr. John Giardina)

Jul 2022 – Jun 2025

- Conceptualized and developed simulation models to conduct clinical and cost-effectiveness analyses of infectious diseases, such as COVID-19 and malaria, for international travelers under the Global TravEpiNet consortium in collaboration with CDC's National Center for Emerging and Zoonotic Infectious Diseases.
- Synthesized literature on simulation models and cost-effectiveness analyses to inform best practices in travel medicine.
- Streamlined and integrated analysis workflows of the Cost-Effectiveness of Preventing AIDS Complications (CEPAC) model with R to enhance research efficiency and reproducibility.
- Conducted training sessions on basic operations and data visualization with R for 40 newly hired research assistants.

International Food Policy Research Institute

Remote

Short-term Research Collaborator (supervised by Dr. Aulo Gelli)

Jul 2022 – Nov 2022

- Analyzed panel survey data of 1,000+ households in Burkina Faso using Stata and R to assess the impact of nutrition-sensitive randomized controlled interventions on livestock mortality. Estimated gender-based investment effects and zoonoses risks with ANCOVA regression, in collaboration with a French-speaking local research partner.

Center for Effective Global Action, University of California, Berkeley

Berkeley, CA

Graduate Student Researcher (supervised by Prof. Edward Miguel)

May 2021 – May 2022

- Performed cost-benefit analyses of deworming randomized controlled interventions in Kenya, evaluating intergenerational child mortality reduction through cost per DALY averted, using Global Burden of Disease data and willingness-to-pay survey from 753 Kenyan respondents.
- Integrated research findings into a web-based cost-benefit analysis application using R, enabling worldwide policymakers to access transparent and reproducible results.

Accenture Japan

Tokyo, Japan

Consultant, Health & Public Service

Dec 2018 – Aug 2020

Analyst, Health & Public Service

Nov 2017 – Nov 2018

- Delivered policy and strategy projects for Japanese national and metropolitan governments, spanning global market analysis, digital public infrastructure (smart cities and COVID-19 risk mitigation), foreign direct investment promotion in healthcare and advanced technologies, and large-scale economic analysis for government white papers.

Ministry of Agriculture, Forestry and Fisheries

Tokyo, Japan

Project Coordinator, International Affairs, Animal Health Div.

Apr 2017 – Oct 2017

Policy Analyst, Livestock Production Promotion Div.

Apr 2016 – Mar 2017

- Supported international animal health and livestock policy, including trade negotiations following BSE outbreak, on-site response to highly pathogenic avian influenza outbreaks, and comparative policy analysis to strengthen biosecurity, livestock productivity, and international market access.

EDUCATION

University of Washington, The Comparative Health Outcomes, Policy, and Economics (CHOICE)
Institute, School of Pharmacy **Seattle, WA**
PhD in Health Economics & Outcomes Research **Sep 2025 – Present**

Academic Advisor: Prof. Jing Li

Related Projects: Cost-effectiveness analysis of GLP-1 RA to prevent type 2 diabetes among teenagers in the United States; Cost-effectiveness and value of information analysis of childhood vaccination against mpox

Teaching Assistant: “Regression methods for health care cost and resource utilization” (*Prof. Anirban Basu & Prof. Noemi Kreif*)

University of California, Berkeley, Goldman School of Public Policy **Berkeley, CA**
Master of Development Practice **May 2022**

GPA: 3.96/4.00

Certificates: Global Public Health; Applied Data Analysis; Geographic Information Science and Technology

Capstone Project: “Data, Machine Learning, and Causal Inference: Real-world Application to Global Health” (*advised by Prof. Alan Hubbard; client: Causal Design*): Applied advanced statistical modeling and machine learning methods to evaluate causal effects of social capital on sanitation outcomes of 600 households in rural Cambodia with R, identifying key social factors impacting hygiene behaviors.

The University of Tokyo **Tokyo, Japan**
Bachelor of Veterinary Science (Doctor of Veterinary Medicine equivalent) **Mar 2016**

GPA: 3.82/4.00 (evaluated by World Education Service)

Concentrations: Law & Politics (2010–2012); Veterinary Medicine (2012–2016)

Thesis: “Pathophysiology of Asian Velogenic Newcastle Disease Virus”: Conducted study of zoonotic virus isolated in Indonesia. Suggested virus kinetic mechanism through statistical analysis of the relationships among infected virus proliferation, antibody generation, and disease progression from 100 chickens’ 11 organ samples.

Research Group: Global Animal Resource Science Laboratory with members from the U.S., China, Vietnam, Afghanistan, and Myanmar (seminars conducted in English)

RELATED COURSEWORK

Society for Medical Decision Making **Oct 2024**

Advanced Discrete-Event Simulations in R

Advanced Computation of Value of Information Measure to Determine Optimal Research Design

Harvard T.H. Chan School of Public Health **Jun – Aug 2022, Jul – Aug 2023**

Global Infectious Diseases Summer Program (Department of Immunology and Infectious Diseases)

Causal Inference: Learn from the Experts (Department of Epidemiology)

Massachusetts Institute of Technology (Online) **May 2019 – Jan 2021, May – Aug 2025**

MicroMasters Program in Data, Economics, and Development Policy: Microeconomics; Designing and Running Randomized Evaluations; Data Analysis for Social Scientists; Probability: The Science of Uncertainty and Data; Evaluating Social Programs

RESEARCH EXPERIENCE

Journal Articles:

- [Accepted] E.P. Hyle, K. Ravishankar, K.E. Mace, **S. Koiso**, A.D. Ridpath, E. Gulbas, J. Thwing, R.C. LaRocque, S. Williams, E.T. Ryan, T.S. Brown. Mapping hospitals that treat severe malaria cases in the United States: Evidence to inform health systems strategies for intravenous artesunate supply. *Clinical Infectious Diseases*, 2026. doi:10.1093/cid/ciag139
- **S. Koiso**, H. Lee, A.L. Ciaranello, M.S. Jalali, K.A. Freedberg, E.T. Ryan, R.C. LaRocque, E.P. Hyle. Effect of containment strategies against respiratory diseases on infections imported via international travel to the USA: a modeling study. *BMJ Open*, 2026;16:e105013. doi:10.1136/bmjopen-2025-105013
- T. Stanic*, **S. Koiso***, N.F. Fields, A.T. Walker, N.M. Mulroy, E.T. Ryan, R.C. LaRocque, E.P. Hyle. Economic value of pre-travel health interventions for communicable diseases in international travellers. *Journal of Travel Medicine*, 2025;32(7):taaf053. doi:10.1093/jtm/taaf053 *: Co-first authors
- **S. Koiso**, E. Gulbas, L. Dike, N.M. Mulroy, A.L. Ciaranello, K.A. Freedberg, M.S. Jalali, E.T. Ryan, R.C. LaRocque, A.T. Walker, E.P. Hyle. Modeling approaches to inform travel-related policies for COVID-19 containment: a scoping review and future directions. *Travel Medicine and Infectious Disease*, 2024;62:102730. doi:10.1016/j.tmaid.2024.102730
- L. Kakoullis, R. Steffen, M. Goeijenbier, A. Osterhaus, S.R. Rao, **S. Koiso**, E.P. Hyle, E.T. Ryan, R.C. LaRocque, L.H. Chen. Influenza: seasonality and travel-related considerations. *Journal of Travel Medicine*, 2023;30(5):taad102. doi:10.1093/jtm/taad102

Working Papers:

- E.P. Hyle, L. Ang, G. Luu, P. Kasaie, D. Dai, **S. Koiso**, J. Phelan, F. Ebem, C. Duggan, E. Humes, P.E. Sax, L. Gerace, J. Giardina, E.J. Orav, A.M. Neilan, A. Pandya, J.F. Figueroa, K.N. Althoff, K.A. Freedberg. Costs associated with increasing numbers of Medicare beneficiaries with HIV aged 65 years and older from 2026 to 2035. *medRxiv*. 2025. doi:10.64898/2025.12.19.25342703
- M. Walker, A. Huang, S. Asman, S. Baird, L. Fernald, J. Hamory, F.H. de la Guardia, **S. Koiso**, M. Kremer, M. Krupoff, M. Layvant, E. Ochieng, P. Suri, E. Miguel. Intergenerational Child Mortality Impacts of Deworming: Experimental Evidence from Two Decades of the Kenya Life Panel Survey. *NBER working paper, no. 31162*. 2023. [Currently under review by *American Economic Journal: Applied Economics*]

Conference Presentations:

- **S. Koiso**. Cost-effectiveness Analysis of GLP-1 RA to Prevent Type 2 Diabetes among Teenagers in the United States.
 - Poster presentation at International Society for Pharmacoeconomics and Outcomes Research (ISPOR) 2026; May 2026 (Scheduled); Philadelphia, PA, USA.
- **S. Koiso**, T. Stanic, N.F. Fields, N.M. Mulroy, E.T. Ryan, R.C. LaRocque, E.P. Hyle. Clinical and economic value of pre-travel health interventions for communicable diseases: a scoping review (1988-2023).
 - Poster presentation at The 2025 Celebration of Science; April 2025; Boston, MA, USA.
- **S. Koiso**, H. Lee, A.T. Walker, A.L. Ciaranello, J. Thwing, M.S. Jalali, K.A. Freedberg, E.T. Ryan, R.C. LaRocque, E.P. Hyle. Modeling the effect of COVID-19 containment strategies on new infections via international travel to the U.S.
 - Poster presentation at 2025 UWSOP Corporate Advisory Board Annual Meeting; September 2025; Seattle, WA, USA.
 - Oral and poster presentation at The 9th Japan-US Science Forum in Boston; November 2024; Boston, MA, USA. (presentation conducted in English)
 - Poster presentation at Society for Medical Decision Making (SMDM) 46th Annual Meeting; October 2024; Boston, MA, USA.
 - Poster presentation at Mass General Research Institute Clinical Research Day 2024; October 2024; Virtual conference.
- T.S. Brown, K.E. Mace, **S. Koiso**, A.D. Ridpath, E. Gulbas, A.T. Walker, R.C. LaRocque, S. Williams, E.T. Ryan, E.P. Hyle. Geographic distribution of hospitalized severe malaria cases in the United States, 2012–2018.
 - Oral presentation at IDWeek 2024; October 2024; Los Angeles, CA, USA.
- **S. Koiso**, E. Gulbas, L. Dike, N.M. Mulroy, A.L. Ciaranello, K.A. Freedberg, M.S. Jalali, A.T. Walker, E.T. Ryan, R.C. LaRocque, E.P. Hyle. COVID-19 modeling approaches for travel-related policies: review and future directions.

- Poster presentation at the 18th Conference of the International Society of Travel Medicine (CISTM18); May 2023; Basel, Switzerland.
- Poster presentation at 73rd MGH Scientific Advisory Committee (SAC) meeting; March 2023; Virtual conference.

Other Works:

- **Book chapter:** S. Koiso. (2026). Pandemic Responses in Low- and Middle-Income Countries. In: Bennett, G., Goodall, E. (eds) The Palgrave Encyclopedia of Disability. Palgrave Macmillan, Cham. doi:10.1007/978-3-031-40858-8_476-1
- **Editorial:** S. Koiso, E.P. Hyle. Cost-Effectiveness of the 2023 to 2024 COVID-19 Booster Vaccine—Reducing Barriers to Access. *JAMA Network Open*. 2025 Aug 1;8(7):e2523695 doi:10.1001/jamanetworkopen.2025.23695
- **Web-based application:** Center for Effective Global Action, Berkeley Initiative for Transparency in the Social Sciences. Open Policy Analysis for Deworming Interventions. 2021: The latest version can be accessed via <https://github.com/BITSS-OPA/opa-deworming>
- **Report:** Accenture Japan Ltd. *Heisei 29 nendo Waga kuni chusho kigyo no kozo bunseki oyobi kozo henka no shorai suikei ni kakawaru itaku jigyo chosa hokokusho* [Fiscal 2017 Commission for Structural Analysis of SMEs in Japan and Projections of Structural Changes Report (in English)]. White Paper on Small and Medium Enterprises (SMEs) in Japan 2018. National Association of Small and Medium Enterprise Promotion Organizations. 57-109; 446-448. 2018: English version can be accessed via https://www.chusho.meti.go.jp/pamflet/hakusyo/H30/PDF/2018hakusho_eng.pdf

LECTURES/TALKS

- “Decision Science: Policy Design and Clinical Decision-Making Through Health Economics” at Boston Japanese Researchers Forum; April 2025; Boston, USA (conducted in Japanese)
- “R Data Visualization” at R Foundations and Introduction to R-HTA Modelling; R-HTA in LMICs; April 2025; Virtual

FELLOWSHIPS/GRANTS/AWARDS

- [Awarded] Higashi Fellowship Travel Funds, 2026
- 2025-2026 Rubenstein Graduate Student Fellowship, The CHOICE Institute UW, USD1,875
- Scholarships for Mid- to Long-Term Studies, Friend of UTokyo, Inc., 2025, USD10,000
- MDP Department Fellowship, UC Berkeley, 2020–2022, USD13,500
- Heiwa Nakajima Foundation Fellowship, Heiwa Nakajima Foundation, 2020–2022, JPY5,600,000 (2021USD approx. 52,000)
- [Awarded but voluntarily declined] Overseas Study Support Program (Graduate Degree Track), Japan Student Services Organization, 2020, JPY4,325,000 (2020USD approx. 40,500)
- Tuition Fee Exemptions, the University of Tokyo, 2014–2016, JPY1,200,000 (2015USD approx. 10,000)
- Student Exchange Support Program (Tropical Veterinary Medicine Training Courses in Asia), Japan Student Services Organization, 2014, JPY70,000 (2014USD approx. 670)

PROFESSIONAL/ACADEMIC/COMMUNITY ACTIVITIES

- Veterinarian, certified by Ministry of Agriculture, Forestry, and Fisheries (May 2016)
- **Member:** Models of Infectious Disease Agent Study (Nov 2024–present); Society for Medical Decision Making (Aug 2024–present); International Society for Pharmacoeconomics and Outcomes Research (Feb 2026–present)
- **Ad-hoc peer reviewer:** *International Journal of Infectious Diseases*, *BMC Public Health*
- **Abstract reviewer:** Society for Medical Decision Making (SMDM) 48th annual meeting.
- Analyzed one health issues and policy: a three-week visit to Vietnam’s FAO country office, slaughterhouses, and live poultry markets (Sep 2013); a two-week training at Thailand’s Kasetsart University (Sep 2014); one-week internship at World Organization for Animal Health (OIE) on international conference on highly pathogenic avian flu (Aug 2014).
- **Volunteer & Community Service:** Missionaries of Charity orphanage in the Philippines: volunteered for disability childcare (Mar 2011); Office for International Cooperation and Exchange (OICE): organized cultural

exchange events and supported international students with studies and laboratory training (Apr 2012–Mar 2016); Universal Peace Federation in Kenya: constructed an official website and trained local staff (Jun–Dec 2019); Kokuren (United Nations) Forum, Internship and Volunteer Group: interviewed Japanese who interned or volunteered in UN organizations (Apr 2019–Dec 2021); Students of Color in Public Policy (SCiPP), UC Berkeley (Oct–Dec 2021); Japan Festival Boston, Culture & Workshop Team, Deputy Leader (Oct 2023–Apr 2025); TransHarvard, Harvard Trans+ Community Celebration (Nov 2024)

- **Leadership:** Accenture Inclusion & Diversity Committee: wrote articles on LGBTQ+ issues for internal monthly journals; joined panel for Work with Pride event for over 100 HR delegates of Japanese companies; organized events for 50 members, inviting guest speakers (Oct 2018–Aug 2020); R for Health Technology Assessment (HTA) in Low- and Middle-Income Countries chapter, Scientific Committee member: organizing and delivering workshops on health economics and decision science using R to empower analysts and healthcare institutions in low- and middle-income countries (Nov 2024–present)
- **Training:** Graduate Student Inclusivity Training, UC Berkeley (Feb–Jun 2021)

SKILLS/LANGUAGES/INTERESTS

- **Computer Skills:** Microsoft Word, Excel, & PowerPoint (expert), R (expert), Python (proficient), ArcGIS (proficient), Github (proficient), LATEX (proficient), TreeAge Pro (basic), Stata (basic), SQL (basic)
- **Language:** Japanese (native), English (fluent), French (proficient): Diplôme d'Aptitude Pratique au Français (APEF) Niveau 2 (Jul 2012), Diplôme d'études en langue française (DELFI) A2 (Feb 2017)
- Avid cyclist: Biked Route 66 from Oklahoma City to Los Angeles (Mar 2016) and from Oklahoma City to Chicago (May 2025); raised \$2,000 to support family and community health efforts for the Centre for Family Health Initiative in Nigeria.